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Exam 4: NE533: Nuclear Fuel Performance
Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

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1. (18 pts) A ZIRLO cladding tube is in reactor at 625 K for 500 days. The initial wall thickness is 600 μm.
   a) Estimate the oxide thickness after this time?
   b) Assuming the hydrogen pickup fraction is 14%, what is the weight ppm of hydrogen in the cladding? Assume PBR = 1.56, ρ_{Zr} = 6.5 g/cc, ρ_{ZrO2} = 5.68 g/cc.
2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?
3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?
4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.
5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?
6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?
7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?
8. (6 pts) List one near-term ATF concept and why it is of interest.
9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?
10. (6 pts) List three examples of limiting phenomena governing LWR operation.
11. (6 pts) What fuel performance and safety impacts does CRUD have?
12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.
13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.

1) $t^*(d) = 6.62\times10^{-7}\exp\left(\frac{11949}{T}\right) = 133.01 \text{ days} < 500 \text{ days}$

$T = 625\,K$

a)

$\delta^*(\mu m) = 5.1\exp\left(\frac{-550}{T}\right) = 2.115\ \mu m$

$K_L\left(\frac{\mu m}{d}\right) = 7.48\times10^{6}\exp\left(\frac{-12500}{T}\right) = 0.0154\ \frac{\mu m}{d}$

18/18

$\delta(\mu m) = 2.115 + 0.0154(500-133.01) = \boxed{7.773\ \mu m}$

b)

$$C_H = \frac{2f \times \delta \times \rho_{ox} \times f^{0}_{ZrO_2} \times \frac{M_H}{M_O} \times 10^6}{\left(t - \frac{\delta}{PBR}\right) \times \rho_{Zr}}$$

$$\frac{2(0.14)(7.773)(5.68)\left(\frac{32}{32+91}\right)\left(\frac{1}{16}\right)10^6}{\left(600 - \frac{7.773}{1.56}\right)(6.5)} = \boxed{51.976 \text{ wt.ppm}}$$

2) Diffusion of oxygen into the cladding.

2/2

3) Hydrides tend to form on the outer radius of the cladding. This is due to the soret effect where hydrogen migrates towards lower temps. Hydrides tend to form at high tensile stress regions which occurs at the outer radius. Hydrides impact thermal conductivity (decrease overall) and reduce ductility, increasing risk of failure.

8/8

4) RIA (reactivity induced accident) occurs when there is a large, drastic spike in reactivity within the core. Typical RIAs are rod ejections (PWR) or rod drop (BWR).

10/12 In both cases, a rod or rod bank is lost causing reactivity to increase. More neutrons are in the system causing fissions and energy to rise. Temperature in the fuel sharply increases which transfers to the cladding. This can lead to PCCI or DNB which can cause a cladding failure. -needed more

5) LOCA (Loss of cooling accident) occurs when a break in the reactor cooling system happens. This leads to a

12/12 temperature increase (slower than RIA) and drop in pressure. These changes promote oxidation and hydride growth, placing the cladding at risk of ballooning/bursting. Additionally, ECCS will switch on, which can rapidly quench the high temp cladding and cause failure. As mentioned, RIA temp rise is faster than LOCA, however, LOCA also has a drop in pressure.

6) Burnup increases failure probability and impacts the type of failure during an accident by changing the microstructure. Higher burnup fuels are more likely

4/4 to have fission products that decrease thermal conductivity or increase corrosion (ex cladding). Both of these put the fuel system at risk of failure.

-cladding brittle

Ways of accident tolerence

7) Increase melting temperature

Increase thermal conductivity properties

4/6 Increase time to failure/ruptre - related concepts, but related to other things

Path ways → changes to the fuel, ~~clad~~ cladding, gap or operating procedures

8) One near term ATF concept is cladding coating (ex ceramic) These coatings do not degrade thermal conductivity or act as a neutron poison as much as changing the cladding entirely. These coatings reduce the cladding contact with the coolant and significantly decrease corrosion.

6/6 -high T Steam Specifically

9) Zr exposed to high temp steam will rapidly oxidize as the high temperatures promote oxygen diffusion into the cladding.

4/6 -needed more

10) Fuel melt

6/6 Cladding wear

PCC/MI

11) CRUD forms an oxide/~~structural~~ material layer on the outside of the cladding. This layer decreases thermal conductivity and promotes corrosion of the cladding (structural failure risk). Additionally, boron can be trapped causing rapid shifts in power. Parts of the now irradiated CRUD can flake off and expose personnel to radiation in areas outside of the reactor.

6/6

12) Switching ~~from~~ from a normal water chemistry to <u>hydrogen water chemistry</u>. This allows for changes in the pH and decreases corrosion. pH has risen to around 7.3-7.4 with this change. Corrosion of Fe and Zr are affected by the pH level so controlling pH is important to reduce it.

6/6

Adding <u>Zinc</u> to the water is done to reduce radiation risks. This happens as Zn infiltrates the CRUD and takes up spots otherwise occupied by CoCO. This means it stays in the water longer to be picked up by the CVCS. Zn is also noble increasing stability.

13)

Columnar

Collander grain growth. Gaseous ~~pores~~ pores with a decreased thermal conductivity cause an evaporation and condensation effect that allow the pores to travel and form long continuous grains near the middle of the max pellet.

6/8

Central Void formation. The pores travel to the center due to the same evaporation and condensation forming a large central void.

-this is two parts of same effect